of choice for domestic and international professional education for the mining industry.

Graduate Certificate in Tailings Engineering

Tailings Engineering specialty is for those who wish to specialize in interdisciplinary fields that include understanding emerging technical and social issues in this discipline. This specialty is open to students with mining or non-mining engineering undergraduate degrees who are interested in scholarship and research on topics including, but not limited to, geology, geotechnical engineering, water, soil mechanics, tailing valorization, mineral processing and extractive metallurgy, environmental engineering, and social governance.

Because of the interdisciplinary nature of this degree program, students will be required to take two required courses in the Department of Mining Engineering and then choose two courses related to their area of interest offered by mining, as well as other departments across campus. The certificate program can be completed by taking 12 credits of graduate coursework.

REQUIRED COURSES:

MNGN581	FUNDAMENTALS OF TAILINGS ENGINEERING I	3.0
MNGN582	FUNDAMENTALS OF TAILINGS ENGINEERING II	3.0

Electives

Please select two courses from the following list of Elective Clusters:

Mining/Mineral Processing & Extractive Metallurgy		
MNGN532	PARTICULATE MATERIAL PROCESSING I - COMMINUTION AND PHYSICAL SEPARATIONS	3.0
MNGN541	ELECTROMETALLURGY	3.0
MNGN542	HYDROMETALLURGY	3.0
MNGN563	WATER WASTE AND MINE CLOSURE	3.0
MNGN556	MINE WATER AND ENVIRONMENT	3.0
Civil & Environmental Engineering		
CEEN515	HILLSLOPE HYDROLOGY AND STABILITY	3.0
CEEN519	RISK ASSESSMENT IN GEOTECHNICAL ENGINEERING	3.0
CEEN581	WATERSHED SYSTEMS MODELING	3.0
CEEN573	RECLAMATION OF DISTURBED LANDS	3.0
Geochemistry		
CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0
CHGC504	METHODS IN GEOCHEMISTRY	3.0
CHGC508	ANALYTICAL GEOCHEMISTRY	3.0
CHGC509	INTRODUCTION TO AQUEOUS GEOCHEMISTRY	3.0
Geophysics		
GPGN530	APPLIED GEOPHYSICS	3.0
GPGN570	APPLICATIONS OF SATELLITE REMOTE SENSING	3.0
ESG		
MNGN503	MINING TECHNOLOGY FOR SUSTAINABLE DEVELOPMENT	3.0
MNGN562	MINING ENVIRONMENTAL AND SOCIAL RESPONSIBILITY	2.0
MNGN567	SUSTAINABLE DEVELOPMENT AND EARTH RESOURCES	3.0
MNGN571	ENERGY, NATURAL RESOURCES, AND SOCIETY	3.0
Geological Engineering		
GEGN542	ADVANCED DIGITAL TERRAIN ANALYSIS	3.0
GEGN573	GEOLOGICAL ENGINEERING SITE INVESTIGATION	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN580	APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES	3.0
GEGN582	INTEGRATED SURFACE WATER HYDROLOGY	3.0

Course substitutions require the approval of the Tailings Engineering graduate certificate program director.

Courses

MNGN501. REGULATORY MINING LAWS AND CONTRACTS. 3.0 Semester Hrs.
(I) Basic fundamentals of engineering law, regulations of federal and state laws pertaining to the mineral industry and environment control. Basic concepts of mining contracts. Offered in even numbered years. Prerequisite: Senior or graduate status. 3 hours lecture; 3 semester hours. Offered in even years.

MNGN502. GEOSPATIAL BIG DATA ANALYTICS. 3.0 Semester Hrs.
Spatial data models (vector and raster data) and structures (R tree and octree data structures), characteristics of geospatial big data (e.g. satellite images, Lidar point clouds, sensor measurements, environmental monitoring data, socio-economic data), geospatial big data sources (IoT, sensors, images, Lidar, crowd-sources), geospatial big data life cycle, visualizations for geospatial big data sets, isual design principles (Bertin`s visual variables, preattentive attributes, Gestalt principles and Tufte`s design principles), the first order and second order exploration methods for various geospatial data (spatially discrete point data, spatially continuous point data and area data), machine learning algorithms (k-means clustering, self-organizing maps, support vector machines), statistical learning methods (point pattern analyses, kriging, non-spatial, spatial regression and geographically weighted regression).
Course Learning Outcomes

- Recognize main features of spatial data models (vector and raster data) and structures (R tree and octree data structures), recall the characteristics of geospatial big data and evaluate and compare various data sets (e.g. satellite images, Lidar point clouds, sensor measurements, environmental monitoring data, socio-economic data) in terms of the 5V`s (Volume, Velocity, Veracity, Variety and Value) of big data.
- Distinguish types of geospatial big data and its sources (IoT, sensors, images, Lidar, crowd-sources, etc.) in geosciences
- Apply incremental steps of geospatial big data life cycle to a given business case (e.g., monitoring land use and land cover change, decision making processes in geosciences, monitoring human behavior, etc.)
- Design at least three data visualizations for geospatial big data sets considering visual design principles (Bertin`s visual variables, preattentive attributes, Gestalt principles and Tufte`s design principles).